LB Maths Tuition Platform

Administrator & Head Tutor Operating Manual

1. Executive Overview & Purpose

The LB Maths Tuition Platform (TutorHub) is a centralized student portal and administrative engine designed to schedule, deliver, and track online mathematics tutoring sessions conducted via Microsoft Teams. As an Administrator (Head Tutor), you possess full operational control over student enrollments, tutor rosters, lesson timetables, conflict resolution, payment tracking, and global security policies.

Portal Area	URL Route	Key Operational Duties
Admin Portal	/admin	Master dashboard: scheduling, rosters, revenue analytics, audit logs, and settings.
Tutor Dashboard	/tutor	Tutor view: assigned student roster, upcoming sessions, delay alerts (+5m/+10m), session notes.
Student Lobby	/ or /student	Student entry: 4-digit PIN / Magic Link login, live lesson countdown, Teams join button.

2. Lesson Scheduling & Timetable Controls

2.1 Creating Lessons & Conflict Detection

- Click **'Schedule Lesson'** in the Timetable tab. Select the tutor, student, date, and times.
  - **Automated Conflict Detector:** The engine automatically cross-checks both tutor and student schedules. If a double-booking or unavailability overlap exists, the system warns the admin before saving.
  - **Recurring Bookings:** Check *'Repeat Weekly'* to schedule up to 12 weeks of repeating sessions in a single click.

2.2 Teams Meeting Link & Safeguarding Lock

- **Teams URL Validation:** The platform strictly validates Microsoft Teams meeting links to ensure students are directed to legitimate calls.
  - **Early Entry Safeguard (5-Minute Window):** Students can enter the lobby 10 minutes beforehand, but the *'Join Teams Call'* button is locked until **5 minutes before** start time. This prevents students from lingering unsupervised in meeting rooms.
  - **Live Delay Broadcasts:** If running behind, tutors or admins can post a **+5 min** or **+10 min** delay alert with an optional reason. The student's lobby updates instantly via Server-Sent Events (SSE) with sound cues.

Safeguarding Policy Notice: Never share meeting links externally. All student sessions must have an assigned tutor. If a lesson must be completed before its scheduled start time, the admin panel requires explicit confirmation to prevent accidental records.

3. Student Roster & Credential Management

- **Enrolling Students:** Navigate to the *Students* tab and click *'Add Student'*. Assign their normal tutor, optional parent contact email/phone, and billing rates.
  - **4-Digit Student PINs:** Students authenticate by selecting their profile and entering their 4-digit PIN. The system enforces strict dual-tier rate limiting: **5 incorrect attempts triggers a 15-minute security lockout**. Admins can unlock locked accounts instantly via the *'Unlock PIN'* button.
  - **Secure Magic Links:** Each student has a unique magic link (e.g., https://portal.lbmathstuition.co.uk/student?key=STU-NAME-HEX) for one-click passwordless access on tablets or phones. Click *'Cycle Magic Link'* to regenerate this key if compromised.
  - **Student Inactivity (Breaks/Holidays):** Toggle a student to *'Inactive'* when they take a term break. Their historical records remain preserved, but they are filtered out of active scheduling lists.

4. Tutor Management & Staff Access

- **Tutor Accounts:** Tutors log in at /login using their registered email and a secure password (minimum 8 characters, hashed with bcrypt).
  - **Password Resets & Token Invalidation:** Updating a tutor's password in the Admin tab automatically increments their tokenVersion, instantly terminating any active sessions on lost or unauthorized devices.
  - **Unavailability Records:** Tutors can mark *'Busy'* hours or *'Holiday'* date ranges in their calendar, preventing admins from scheduling lessons during those periods.

5. Rates, Billing & Revenue Analytics

The platform includes a dedicated **Revenue Analytics Hub** in the Admin Portal for tracking tuition turnover, tutor payroll, and profit margins.

Feature	Mechanism	Operational Impact
Student Fee	studentPay (per hour)	Strictly confidential to Head Tutor. Tutors and students can never view this field.
Tutor Pay	tutorPay (per hour)	Visible to the assigned tutor on their dashboard for their own sessions only.
Historical Rate Snapshots	Immutable Archival	When a lesson completes or archives, its fee and payout are locked. Future rate changes never alter past financial history.
Billing Duration Mode	Proportional vs. Round to Hour	Configured in Settings: calculate pay based on exact minutes taught or round to nearest whole hour.
CSV Exports	One-Click Export	Export filtered lessons, dates, and payout figures into Excel/CSV for accounting and payroll.

6. Built-in Classroom Tools & Accessibility

- **Virtual Casio fx-83GT X ClassWiz Calculator:** Full-fidelity scientific calculator with Natural V.P.A.M. exact math engine (fractions, exact π, surds). Students and tutors can pop this up directly inside the app.
  - **Formula Sheet Hub:** Integrated GCSE (Edexcel, AQA) and A-Level (Edexcel, AQA, OCR) exam aids and formula sheets accessible with one click.
  - **Shared Resources Hub:** Admin can upload and tag worksheets, past paper links, and video resources filtered by topic (Algebra, Geometry, Calculus, Statistics).
  - **Sensory & Accessibility Preferences:** Includes high-contrast toggle, OpenDyslexic font support, reduced motion, and audio cue toggles.

7. Audit Logs & System Backups

- **Full Audit Trail:** Every lesson creation, cancellation, reschedule, and rating edit is recorded with the actor ID and timestamp under the *Activity Logs* tab.
  - **Encrypted Database Backups:** In the *Settings* tab, admins can trigger a complete JSON backup export of all system tables (users, sessions, settings, resources). All sensitive password hashes are automatically scrubbed from backup files.
  - **Server Synchronization (SSE):** All timetable changes, delay notices, and online presence indicators synchronize live across all connected devices without manual page refreshing.

Support & Maintenance:
- Production Domain: https://portal.lbmathstuition.co.uk
- Edge Host: Deno Deploy (with LibSQL edge SQLite database on Turso)
- Technical Issues: Contact Platform Administrator (Luke) or reference internal platform docs in /docs.